The Natural Geometry Hierarchy: A Synthesis of the GeoVac Quantum-Chemistry Arc*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*
(Dated: August 14, 2026)

The Geometric Vacuum (GeoVac) framework discretizes non-relativistic electronic structure on *natural geometries*: for each separable quantum system there is a coordinate system in which the Coulomb singularities become boundary conditions and the Laplace–Beltrami operator discretizes on a sparse graph with quantum-number labels for nodes. This synthesis assembles the framework's quantum-chemistry arc—the twelve Group 2 papers spanning one-electron atoms, the two-center one-electron ion, the two-electron atom, the two-center two-electron molecule, the core–valence and polyatomic extensions, the genuine two-center tensor census, the transcendence frontier at the third center, and the isoenergetic Sturmian secular equation—into a single coherent narrative organized by the *natural geometry hierarchy.* The hierarchy is a grid indexed by the number of nuclear centers and the number of electrons: S^3 (Fock) for atoms [21], the prolate spheroid for H_2^+ [23], hyperspherical coordinates for He [25], molecule-frame hyperspherical coordinates for H_2 [26], and composed fiber bundles for core–valence and polyatomic systems [27, 29]. Three threads run through the arc. *First*, an algebraic-first program: the angular content is computed exactly from Gaunt coefficients and Wigner $3j$ symbols at every level, the prolate-spheroidal electron–electron repulsion is replaced by an algebraic Neumann expansion [24], the radial solvers reduce to three-term Laguerre recurrences, and the split-region Legendre nuclear coupling terminates exactly via the $3j$ triangle inequality [26]; what currently survives as quadrature (Z_{eff} screening, the adiabatic potential curves, the hyper-radial coupling) is named explicitly, and finding algebraic replacements is an ongoing goal rather than a deficiency. *Second*, two structural negative theorems delimit the design space: in the single-n D-matrix *approximation* to a single-center molecular Sturmian basis with a shared momentum scale, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$ and share a single orthogonal $SO(4)$ Wigner-D congruence, so the generalized eigenvalues are independent of the internuclear distance and cannot bind unless the Sturmian scale $\beta_k(R)$ itself becomes R-dependent [22]. (Paper 8's own scope remark is explicit that this delimits that approximation and is *not* a no-go for Sturmian molecular binding: the genuine cross-n Coulomb–Sturmian overlap breaks the shared congruence and supplies the R-dependence algebraically—see the Paper 60 entry below, where the method binds.) And a graph built by concatenating atom-centered lattices has an R-independent intra-atomic kinetic energy, producing a monotonically attractive potential-energy surface with no equilibrium [35]. A later genuine two-center census sharpens where the sparsity lives: evaluated over orbital products that span two nuclei the Gaunt l-selection rule fails and only the axial m-rule survives, so the angular sparsity is an *atomic-sector* property—the abelian residue of the molecular point group [30]. *Third*, honest accuracy ceilings: the framework reaches 0.0002% for H_2^+, 0.022% (properly variational, raw $l_{\max} = 7$) down to 0.004% (a non-variational cusp extrapolation) for He, 96.0% of D_e for H_2, and—in the composed architecture, whose accuracy bottleneck is the Phillips–Kleinman pseudopotential—5.3%, 11.7%, and 19.4% equilibrium-bond-length error for LiH, BeH_2, and H_2O respectively. The classical-solver investigation is complete: its accuracy ceilings are characterized, and the framework is positioned as a zero-parameter, structurally sparse *research instrument*—a computationally principled alternative whose primary downstream value (qubit-Hamiltonian sparsity for quantum simulation) lives in the quantum-computing arc—not as a replacement for production quantum chemistry. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps; the interpretive question of which is more fundamental is left open.

I. INTRODUCTION

The Group 3 foundations arc of GeoVac establishes that the hydrogen atom's momentum-space Schrödinger equation maps, by Fock's 1935 stereographic projection [1], onto the free Laplacian on the unit three-sphere S^3, and that a discrete graph on quantum-number labels converges to the S^3 Laplace–Beltrami operator in the continuum limit [19–21]. That arc answers a question about a single one-electron atom. The present synthesis takes up the question the foundations arc opens but does not close: what is the right discretization for a *molecule*, or for an atom with *more than one electron*?

The organizing answer is the *natural geometry principle*, stated first in Paper 11 [23]: every separable quantum system has a natural lattice, namely the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where the problem separates. For a one-center

* Group 2 synthesis in the Geometric Vacuum series

one-electron system that coordinate system is S^3; for a two-center one-electron system it is the prolate spheroid; for a one-center two-electron system it is the hyperspherical coordinate system; for a two-center two-electron system it is the molecule-frame hyperspherical system; and for systems with electrons on separated length scales (a tight core and a diffuse valence shell) the natural object is a *composed* geometry—a fiber bundle in which each electron group inhabits its own natural coordinate system, coupled by screening functions and pseudopotentials. Crucially, this is not a linear tower in which each level subsumes the previous one: the Level-4 two-electron coordinates (R_e, α) are simply undefined for a single electron, so the hierarchy is a two-dimensional grid indexed by (number of centers) × (number of electrons) [27].

This grid structure is the reason the chemistry arc is not a single algorithm but a sequence of natural geometries, each with its own separation of variables, its own algebraic content, and its own accuracy ceiling. Table I summarizes the hierarchy and the headline result at each level; the body of this synthesis unpacks each row, the algebraic structure that makes it work, and the two proven theorems that mark the boundaries of what natural geometry can and cannot do.

GeoVac is presented throughout as a discretization framework that exploits natural geometry, not as a new foundation for quantum mechanics. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by the proven conformal maps of the foundations arc, and the interpretive question of ontological priority is left to the reader. The framework's concrete advantages—a zero-parameter construction from nuclear charges and geometry alone, angular-momentum selection rules baked into the basis, exact algebraic matrix elements, and structural sparsity—are independent of that interpretive question and are the properties this synthesis foregrounds.

II. LEVEL 1–2: FROM THE ATOM TO THE DIATOMIC ION

A. The atomic baseline and the bond sphere

The one-electron atomic baseline is the S^3 graph of the foundations arc: the discrete Fock lattice reproduces the Rydberg spectrum and solves one-electron atoms to better than 0.1% with no free parameters [20, 21]. The natural first question for chemistry is whether the same S^3 object handles a *molecule*. Paper 8 [22] pursues this directly: it shows that a diatomic molecule is geometrically a single S^3 with two weighted poles—the *bond sphere*—in which the internuclear distance R encodes as a polar angle γ through $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$, and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. The geometry is consistent, and it yields exact selection rules for σ-bonds: the bond rotation preserves $m = 0$ within $l = 1$ and does not mix s and p states, $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all γ [22].

B. The Sturmian structural theorem (a guardrail)

The bond sphere construction also produces the arc's first proven *negative* theorem, and it is load-bearing for everything that follows. In the molecular Sturmian basis [10] with a shared momentum scale p_0, the one-electron Hamiltonian matrix is

$$H_{ij} = -\frac{p_0^2}{2}\, S_{ij} + (1 - \beta_j)\, V_{ij}, \tag{1}$$

where S_{ij} is the overlap matrix, V_{ij} the nuclear-attraction matrix, and β_j is the Sturmian eigenvalue of orbital j. The Hamiltonian is *not* a scalar multiple of S: both (H, S) carry their entire R-dependence through the same $SO(4)$ Wigner-D congruence $D^{(n)}(\gamma(R))$, so the generalized eigenvalues $\varepsilon_k = \frac{1}{2}p_0^2 - p_0^2/\beta_k$ (which equals $-p_0^2/2$ only in the degenerate case $\beta_k = 1$) are independent of the bond angle γ, hence independent of the internuclear distance R [22]. A basis that produces R-independent eigenvalues cannot describe binding, which is by definition an R-dependent lowering of the energy—so *within this approximation* binding is excluded. Paper 8 sharpens this further: for a heteronuclear diatomic with charge ratio $Z_A/Z_B > 1$ there is no atom-centered shared-p_0 Sturmian basis that binds both centers simultaneously, and the dual-scale repair drives the two p_0 values toward a geometric mean that eliminates the scale separation [22].

This theorem is a guardrail in the project's sense: it rules out an entire class of single-center, shared-exponent molecular encodings *built on the single-n D-matrix approximation* (nested hyperspherical, unified-basis, single-p_0 Sturmian CI), and one resolution—reintroduce the R-dependent spectrum by moving to the prolate spheroidal coordinates where the diatomic actually separates—is the natural-geometry principle in action.

a. Scope correction (Paper 8 `rem:overlap_imposed`*).* The theorem's backing test *imposes* $S_{AB} = f\, D^{(n)}$ by construction, so what it verifies is a property of the bond-sphere D-matrix model, not of the genuine cross-n Coulomb–Sturmian basis (where, e.g., $\langle \chi^A_{1s} | \chi^B_{2s} \rangle = -0.47$ exceeds the within-n diagonal $+0.35$ at $R = 3.015$). A cross-n overlap breaks the shared congruence, so binding is *not* excluded for the genuine basis: Paper 8's `cor:binding` records that the R-dependence is then supplied *algebraically* by the Shibuya–Wulfman overlap (a function of $s = p_0 R$), needing no continuous two-center PDE, and Paper 60 exhibits the binding directly. Prolate spheroidal geometry remains the route this arc took; it is not the only one the theorem permits.

TABLE I. The GeoVac natural geometry hierarchy for quantum chemistry. Each level is the coordinate system in which the corresponding system separates; the result column reports the headline figure at its stated tier, with the source paper. Composed (Level 5) results are limited by the Phillips–Kleinman pseudopotential, the characterized accuracy ceiling of the composed architecture (Sec. V).

Level	System (centers, electrons)	Natural geometry	Headline result
1	atom (1, 1)	S^3 (Fock projection)	$< 0.1\%$ (one-electron)
2	H_2^+ (2, 1)	prolate spheroid	0.0002% energy (spectral Laguerre)
3	He (1, 2)	hyperspherical	0.022% (2D var. raw) to 0.004% (cusp-extrapolated, non-variational);
4	H_2 (2, 2)	molecule-frame hyperspherical	96.0% of D_e ($l_{\max} = 6$, 61 channels)
4N	LiH (2, 4)	full mol-frame ($SO(12)$)	$R_{\text{eq}} \approx 63.5\%$ ($l_{\max} = 2$, 2D variational; unbound D_e, no PK)
5	LiH (core+valence)	composed (L3 $\oplus$ L4)	R_{eq} 5.3% (l-dependent PK)
5	BeH_2 (polyatomic)	composed + 1-RDM exchange	R_{eq} 11.7%
5	H_2O (triatomic)	composed (five-block, uncoupled)	R_{eq} 19.4%

C. The prolate spheroidal lattice for H_2^+

Paper 11 [23] implements that resolution. The two-center one-electron problem separates in prolate spheroidal coordinates (ξ, η, ϕ) [3], and the separated equations discretize on a prolate spheroidal lattice with a spectral angular solver (Legendre basis) and a spectral radial solver [18], with Brent root-finding in the separation parameter $c^2 = -R^2 E_{\text{elec}}/2$ to enforce self-consistency. The method has zero free parameters: all inputs are nuclear charges, the internuclear distance, and the basis resolution. With the spectral Laguerre radial solver at $n_{\text{basis}} = 20$, the H_2^+ ground-state energy is $E_{\min} = -0.6026$ Ha against the exact -0.6026 Ha [2]—a 0.0002% error, the most accurate single result in the chemistry arc—with the correct dissociation to $H + p$ as $R \to \infty$ [23]. The spectral solver achieves a 250× dimension reduction, a 270× speedup, and a 5000× accuracy improvement over a finite-difference baseline whose 1.01% error is shown to be a discretization artifact, not an intrinsic limit of the basis [23].

The algebraic structure here is the first concrete instance of the arc's central theme. For σ states ($m = 0$), the spectral Laguerre matrix elements are computed entirely from three-term recurrence relations on Laguerre moment matrices, eliminating numerical quadrature from the radial solver completely (the overlap is exact, $S_{mn} = \delta_{mn}/2\alpha$, and the kinetic and potential matrices are finite sums of moments, reproduced to $\sim 10^{-12}$ against quadrature) [23]. For π and δ states ($m \neq 0$) an associated Laguerre basis absorbs the centrifugal $1/x$ singularity via a partial-fraction decomposition and a Stieltjes integral recurrence, reducing the non-algebraic content of the entire solver to a *single* transcendental seed, the exponential-integral value $e^a E_1(a)$; all higher moments follow from the rational recurrence $S_0^{(k)}(a) = \Gamma(k) - a\, S_0^{(k-1)}(a)$ for $k \geq 1$ [23]. This is the exchange-constant taxonomy of the foundations arc made concrete: the irreducible transcendental content of the diatomic radial problem is one seed in the *composition* tier, tagged to a specific integral, rather than diffuse numerical error [28].

III. LEVEL 3: THE TWO-ELECTRON ATOM AND THE CUSP

A. Algebraic V_{ee} on the prolate spheroidal lattice

Before leaving the diatomic geometry, Paper 12 [24] adds the second electron and confronts the electron–electron repulsion $\langle \phi_i | 1/r_{12} | \phi_j \rangle$ over the six-dimensional two-particle configuration space. Standard numerical quadrature on a grid converges slowly because of the Coulomb singularity at $r_{12} = 0$ and saturates near 80% of the exact H_2 dissociation energy regardless of basis size. The algebraic replacement is the Neumann expansion [4] of $1/r_{12}$ in prolate spheroidal coordinates, which decomposes each six-dimensional integral into a finite sum of products of one-dimensional auxiliary integrals computable by recurrence relations: the angular moments $C_l(q)$, the exponential moments $A_n(\alpha)$, and the first- and second-kind Legendre moments all satisfy exact three-term recurrences, so the V_{ee} matrix is exact within the Neumann truncation order with no quadrature grids and no fitted parameters [24]. A selection rule $C_l(q) = 0$ for $q < l$ or $(q - l)$ odd truncates the effective sum automatically.

On H_2 at $R = 1.4011$ bohr the algebraic Neumann V_{ee} reaches 92.4% of the exact D_e with 72 basis functions, against 80.1% for numerical V_{ee} at the same basis size—a uniform 12–20 percentage-point improvement at every basis size from 1 to 72 functions, and the Neumann result plateaus, confirming that the V_{ee} *integration* error has been eliminated [24]. The paper then diagnoses the remaining 7.6% gap honestly: it is a one-electron basis completeness limit, not an integration error. The electron–electron cusp requires the non-analytic terms $r_{12}^{1/2}$ and $r_{12} \ln r_{12}$ (the Kato/Fock structure [7]) that no polynomial prolate spheroidal basis can represent. This is the framework's embedding-tier transcendental: the cusp is two-dimensional in the angular pair coordinate and is the canonical example of an exchange constant that no single algebraic insertion reproduces [28]. The diagnosis identifies the next natural geometry—hyperspherical coordinates, where the coalescence is a boundary condition rather than a coordinate singularity.

B. The hyperspherical lattice for helium

Paper 13 [25] introduces that geometry. In hyperspherical coordinates [8, 9] the He problem becomes a coupled-channel expansion in which the angular eigenvalue problem depends parametrically on the hyperradius R, so the channel functions vary with R; this coupled-channel structure has the form of a fiber bundle and is the mathematical prototype for electron–nuclear coupling in molecules. The electron–electron cusp at $\alpha = \pi/4$, $\theta_{12} = 0$ becomes a boundary condition rather than a singularity [25].

The accuracy results here are reported at carefully separated tiers, and the synthesis preserves the distinction. The original single-channel adiabatic solver gives $E = -2.9052$ Ha, nominally 0.05% from the exact -2.9037 Ha, but it is *non-variational*: it lies below the exact energy through a fortuitous cancellation of adiabatic and finite-difference error, and the paper labels it as such [25]. The properly variational results come from a two-dimensional variational solver that treats the hyperradius and hyperangle simultaneously, avoiding the adiabatic approximation: 0.022% raw at $l_{\max} = 7$ (a genuine variational upper bound) and 0.004% with a Schwartz cusp correction at $l_{\max} = 4$ (a non-variational extrapolation that lies marginally below the exact energy) [25]. A separate, fully zero-parameter route is the graph-native configuration interaction, which uses the graph Laplacian as the one-body operator with exact rational Slater integrals and reaches 0.19% at $n_{\max} = 7$ with no free parameters [25]. Each route exercises a different part of the algebraic registry: the angular sector is exact throughout (Gaunt integrals $G(l, k, l') = 2\,(l\,k\,l' \,|\, 0\,0\,0)^2$, $SO(6)$ Casimir eigenvalues $\nu(\nu + 4)/2$, closed-form perturbation coefficients), while the adiabatic potential curves $U_{\text{eff}}(R)$ and the parametric angular eigenvalues $\mu(R)$ remain point-by-point diagonalizations—numerical content named explicitly as a current, not fundamental, feature [25, 28].

Paper 13 also documents an instructive negative result: in the *adiabatic* approximation, increasing $l_{\max}$ beyond 0 worsens the energy, because the finite-difference treatment of the boundary singularities $l(l + 1)/\cos^2\alpha$ and $l(l+1)/\sin^2\alpha$ degrades faster than the partial-wave content improves; the adiabatic channel truncation has an intrinsic floor near 0.19%–0.20%. The 2D variational solver is what breaks that floor [25]. The same fiber-bundle structure, coupled to a nuclear lattice, delivers an end-to-end demonstration: ab initio rovibrational constants for H_2 with zero spectroscopic input, the fundamental $\nu_{01} = 4157$ cm^{-1} (0.1%) and the rotational constant $B_e = 59.5$ cm^{-1} (2.2%) [25].

IV. LEVEL 4: THE TWO-CENTER TWO-ELECTRON MOLECULE

The two-center two-electron molecule H_2 [5] combines all five Coulomb singularities and had resisted a single natural coordinate system. Paper 15 [26] introduces molecule-frame hyperspherical coordinates $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by the dimensionless ratio $\rho = R/(2R_e)$, where R is the internuclear distance and R_e the electronic hyperradius. The electron–electron cusp appears at $\alpha = \pi/4$, $\theta_{12} = 0$ *independent of molecular geometry*—a structural advantage over prolate spheroidal coordinates, where the cusp is a coordinate singularity—and the angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a two-dimensional parameter sweep to one dimension [26].

A coupled-channel expansion in body-frame partial waves (l_1, m_1, l_2, m_2) with the gerade constraint $l_1 + l_2$ even and $m_1 + m_2 = 0$, solved with the same two-dimensional (R_e, α) variational solver that avoids the adiabatic approximation, recovers 96.0% of the exact D_e [6] at $l_{\max} = 6$ with σ and π channels (61 channels; a complete-basis extrapolation reaches $\sim 97\%$) [26]. This exceeds the 92.4% that prolate spheroidal CI with algebraic Neumann integrals achieved (Paper 12), confirming the cusp-resolution advantage of the hyperspherical geometry. The framework extends to heteronuclear diatomics: for HeH^+ a charge-center origin shift and removal of the gerade constraint recover 93.1% of the exact D_e at $l_{\max} = 4$ with $\sigma + \pi$ channels in the *adiabatic* treatment [26]; this is a non-variational over-binding figure—the consistent two-dimensional variational treatment recovers only $\sim 5\%$ of D_e, so this is not parity with the H_2 result.

The algebraic content reaches its cleanest statement here. The nuclear potential coupling is a split-region Legendre expansion whose sum over Gaunt integrals terminates *exactly* at $k = l' + l$ by the $3j$ triangle inequality $|l' - l| \le k \le l' + l$—no truncation parameter, exact to machine precision in all regimes [26]. This is the arc's prime example of a continuous integration replaced by a finite algebraic sum once the right structure is identified. What remains numerical is named: the Z_{eff} screening correction (Gauss–Legendre quadrature), and the angular eigenvalue sweep at each ρ, which is a point-by-point diagonalization because the split-region structure is piecewise smooth [26, 28].

Paper 15 is careful about variational hygiene. The adiabatic approximation systematically *over*-estimates D_e by roughly 11% relative to the 2D variational solver, and at $l_{\max} = 4$ with $m_{\max} = 1$ the adiabatic D_e exceeds the exact value, violating the variational principle; the 2D result is the correct bound, and the ~ 10 percentage-point gap between them is the kinetic-energy cost of non-adiabatic coupling [26]. The convergence also shows an even–odd staircase: even-$l_{\max}$ increments add large gains (the gerade constraint forces direct nuclear coupling only at even $l_{\max}$) while odd increments are negligible—a selection-rule signature, not a numerical accident [26].

V. LEVEL 5: COMPOSED GEOMETRIES AND THE ACCURACY CEILING

A. The composition principle

Systems with core electrons resist a single natural geometry because core and valence electrons occupy different length scales, and—as noted in Sec. I—the hierarchy does not collapse downward (Level-4 coordinates are undefined for one electron) [27]. Paper 17 [27] introduces *composed* natural geometries: fiber bundles in which each electron group inhabits its own natural coordinate system, coupled by screening functions and ab initio Phillips–Kleinman (PK) pseudopotentials [17] derived entirely from the core wavefunction. For LiH, the Li^+ $1s^2$ core is solved by the Level-3 hyperspherical method, yielding a screening function $Z_{\mathrm{eff}}(r)$ and the PK parameters; the two valence electrons are solved by the Level-4 molecule-frame method with Z_{eff} injected; and the potential-energy surface is scanned by a ρ-collapse adiabatic method. The construction uses *zero* experimental molecular data—all PK inputs are atomic (the core wavefunction, the ionization energy, the nuclear charges) [27].

B. Results and the PK accuracy ceiling

The composed LiH potential-energy surface gives $R_{\mathrm{eq}} = 3.21$ bohr at $l_{\max} = 2$ (6.4% error versus the experimental 3.015 bohr [16], improved from 17% in the LCAO baseline) and $\omega_e = 1471$ cm^{-1} (4.6%); an l-dependent PK variant, which restricts the projector to the $l = 0$ core channels, reduces the bond-length error to 5.3% [27]. The same parameter-free formula transfers to BeH_2 and H_2O: with full one-reduced-density-matrix inter-fiber exchange, BeH_2 reaches 11.7% bond-length error (matching a single-parameter fitted exchange model at zero free parameters), and the uncoupled H_2O construction reaches 19.4% (improved from 26% by general solver evolution—*not* by the PK-consistency fix, which the uncoupled PES does not invoke) with a correct qualitative surface [27].

Two findings establish that the PK pseudopotential is the *characterized accuracy ceiling* of the composed architecture, not a tunable knob. First, the PK barrier is essential, not cosmetic: removing it collapses R_{eq} to 0.88 bohr, so PK is what creates the bonding-versus-Pauli-repulsion balance that produces an equilibrium at all [27]. Second, the residual error is structural. Across all four solver×PK combinations the bond-length error saturates near 5–6% at the $l_{\max} = 2$ operating point, and the $l_{\max}$ drift (the systematic outward creep of R_{eq} with partial-wave order) is not removed by the 2D variational solver, by the l-dependent PK, or by an R-dependent PK whose reference length cannot be derived from atomic quantities without circular reasoning [27]. The paper attributes this to a deep correlation wall—labeled W1e, the composed architecture's instance of the framework's multi-focal composition wall pattern—and documents that single-mechanism repairs each saturate well below closure: a bonding-orbital PK extension at a 43% ceiling, explicit-core Hartree pressure at $\sim 26\%$, basis enlargement at $\sim 10\%$, with Schmidt orthogonalization and core correlation each contributing negligibly [27]. This wall is later relocated from the correlation treatment to the Hamiltonian specification itself, and given a group-theoretic mechanism, in Sec. VI.

C. Equilibrium without PK: the Level-4N control

That PK is a *cost of factorization* rather than a missing physics term is established by the Level-4N control, the exact N-electron generalization in which all four LiH electrons share one S^{11} Hilbert space under $SO(12)$ with S_4 antisymmetry [27]. At $l_{\max} = 2$ (12 S_4 channels, angular dimension 2592) the Level-4N surface has a genuine equilibrium near $R_{\mathrm{eq}} \approx 1.0$–1.1 bohr ($\approx 64\%$ error) with *no* PK—demonstrating that binding exists without the pseudopotential, and that the $l_{\max} = 2$ $S_4\,[2,2]$ angular basis is genuinely insufficient for LiH binding rather than the encoding being wrong. The composed geometry trades this exact inter-group antisymmetry for a 144× angular compression (cost scaling $l_{\max}^2$ versus $l_{\max}^{11}$), which is what makes the composed architecture the production operating point despite the 5–6% ceiling [27]. The reason the trade is forced and not a matter of convenience is a structural obstruction: exact antisymmetry requires a shared coordinate system, while composition factorizes the wavefunction into incompatible coordinate systems—so the three tested PK alternatives (node exclusion, a fiber-bundle Berry connection, spectral orthogonality) each fail for the same reason [27].

VI. THE PK-FREE ALTERNATIVE AND THE FCI ENCODINGS

A. Balanced coupled composition (Paper 19)

Paper 19 [29] pursues the natural follow-on: can the PK pseudopotential be replaced by explicit cross-block physics? The balanced coupled Hamiltonian adds cross-center nuclear attraction V_{ne} via a multipole expansion with exact Gaunt termination (the multipole sum terminates at $L_{\max} = 2l_{\max}$ by the same $3j$ selection rules, with no truncation error), restoring the cross-block one-body coupling that the composed architecture lacks; these missing one-body terms are the Shibuya–Wulfman integrals of Coulomb Sturmian theory [11, 29]. The result is the PK-free alternative for single-point energies at fixed geometry: LiH four-electron FCI reaches 0.20% energy error at $n_{\max} = 3$ (down from 1.8% at $n_{\max} = 2$), using analytical incomplete-gamma cross-center integrals at machine precision [29]. The structural cost is geometric rather than energetic: the equilibrium bond length still drifts to $R_{\rm eq} = 3.28$ bohr ($\approx 8.8\%$ error), though the drift per $n_{\max}$ step is about three times smaller than the PK $l_{\max}$ drift [29]. The honest reading is that the energy axis converges through $n_{\max} = 3$ while the geometry drifts—but not that the energy axis is simply the good one: at $n_{\max} = 4$ the model crosses *past* exact (over-binding), so the fixed-geometry error is not monotonically improvable either [29]. The balanced coupled Hamiltonian is therefore suited to single-point quantum simulation at fixed geometries rather than to full potential-energy surfaces, with that non-monotonicity disclosed. Paper 19 also records a clean negative: cross-block electron–electron integrals *without* the matching cross-center V_{ne} are energetically unbalanced (29% error, worse than PK), and a harmonic-driven orthogonalization wall blocks second-row hydrides such as NaH, which PK cross-center coupling reduces but does not close [29]. That NaH wall is crossed in Paper 58 (next subsection), by evaluating the integrals over genuine two-center orbital products rather than atomic-sector surrogates—at the cost of the label-generated sparsity [30].

B. The abelian residue: why the sparsity is atomic-sector (Paper 58)

Paper 58 [30] closes the question the balanced-coupled work leaves open—whether the label-generated sparsity and the genuine cross-center physics can coexist—and the answer is group-theoretic rather than technical. Every prior molecular builder evaluates its integrals in the *atomic sector*: each orbital product entering an integral sits on a single nucleus, and inter-nuclear coupling enters through a surrogate (the composed builder's $Z_{\rm eff}$ screening; the balanced builder's displaced-nucleus one-body term, with both orbitals still on one center). Because neither ever forms a genuine two-center orbital product, the Gaunt l-selection rule is exact for them *by construction*, and the sparsity follows.

Evaluated instead over orbital products that genuinely span two nuclei, $\langle\chi^A|\cdot|\chi^B\rangle$, in exact rational arithmetic (Mulliken auxiliary functions for the overlaps; a hydrogenic eigen-trick for the cross-center one-body term, certified entry-wise by an exact bra/ket identity), the census measures that the Gaunt l-selection rule does *not* survive while the axial m-rule does, and the symmetry-permitted electron-repulsion density inflates 13.8× over the builder at $n_{\max} = 2$. An internal theorem supplies the mechanism. The label m names the abelian symmetry both centers share—rotation about the internuclear axis is a single global one-parameter group—while l names irreducible representations of a rotation group attached to one center. The intersection of the two centers' rotation groups is exactly the axial $U(1)$, whose character group is $\mathbb{Z}$ by Pontryagin duality; abelian selection rules are additive charge conservation and survive, whereas non-abelian Clebsch–Gordan selection needs the full sphere and does not. *The surviving cross-center sparsity is the abelian residue of the molecular point group, readable in advance.*

Two re-readings follow. First, the sparsity previously charged to Löwdin orthogonalization (a 17.9× Pauli inflation) was largely a bill already owed: the genuine tensor's own permitted density inflates by the same order ($\sim$15×) before any transformation, so orthogonalization is not, in the bulk, the cause of the density blamed on it. Second, four earlier relocations of the non-orthogonality each moved the cost without reducing it—the cost measures the physical asymmetry of a two-center problem—and equivalent-atom-swap tapering *reduces* the qubit saving while inflating the Pauli count, because identifying blocks by permutation destroys the per-block abelian labels that generate the sparsity. The sparsity-versus-binding trade is a conservation law, not a defeat.

The corresponding gain is a panel-verified demonstration. All-electron twelve-electron NaH—the framework's hardest documented binding failure, which six Phillips–Kleinman modifications, an explicit-core Hartree–Fock treatment, and a DMRG cross-check had each left unbound—binds at $R_{\rm eq} = 3.736\,a_0$ (+4.8% versus experiment), recovering 91.1% of the in-basis FCI binding from three fragment-native determinants, and with *fewer* correlation degrees of freedom than the runs that failed—which is what carries the inference. This localizes W1e to the Hamiltonian specification rather than the correlation treatment: the wall was the atomic-sector construction, not the determinant expansion. The demonstration deliberately forfeits the sparsity, because the sparsity and the atomic-sector evaluation are the same choice.

A subsequently supplied two-center electron-repulsion engine closes all four integral classes in exact form (at transcendence weight one, π-free over $\{E_1, \ln, \gamma\}$), which upgrades the $(AA|BB)$ block of the census from *counted* to *decided*: all 195 symmetry-permitted entries are proven genuinely nonzero by Lindemann–Weierstrass separation, with no accidental zeros and none missed by the counting. For the three genuinely cross classes—79.5% of the tensor—Paper 58 states a three-rung position that is *materially better than counted*: the seed functions $\{1, e^{-\lambda R}, E_1(\mu R), \ln R\}$ are linearly independent over $\mathbb{Q}(R)$ by Liouville–Rosenlicht, so every permitted entry is *functionally* nonvanishing (unconditional) and generically nonzero (measured); but it is pointwise *decidable* only conditional on a Schanuel-type independence hypothesis for the E_1 and γ values. The block is therefore blocked from DECIDED by open transcendence, *not* by missing labour—E_1-value independence at algebraic points is an open problem and γ is not known even to be irrational [30]. Weight one is not the same as decidable.

C. The transcendence frontier at the third center (Paper 59)

Paper 58 closes the two-center tensor in closed form; Paper 59 asks what happens at the third. Read in momentum space, the three-center electron-repulsion integral is a two-scale Bessel moment whose two Fock scales define a quartic curve: the two-center case is genus zero (elementary, weight-one $\{E_1, \ln, \gamma\}$), and the third center raises the genus to one. Its period is a complete elliptic integral, it satisfies an explicit fourth-order Picard–Fuchs equation, and the connection is *irreducible* (a Fourier–Laplace transform of an irreducible rank-one connection), so no closed form arises through factorization [31]. The collinear observable is certified to 66 digits and guarded integer-relation searches are decisively negative through weight three at clean heights (the companion Paper 61 [32]); the finite closed form remains open, and the paper says so.

Two consequences for this arc. The polyatomic question below is thereby sharpened from "can the genuine-integral route be extended?" to a statement about transcendence class: the extension is not blocked by missing algebra but by a genus jump, and any closed-form polyatomic tensor is all-or-nothing (a part-closed-form, part-fitted tensor is not decidable at all). And the same obstruction gates explicitly-correlated (F12) evaluation, because the difficulty of a two-electron integral lives in the densities, not in the operator between them.

D. The isoenergetic secular equation as a quantum algorithm (Paper 60)

Paper 60 carries the Sturmian thread into the quantum-algorithm setting and, in doing so, exhibits the binding the scoped guardrail above does not forbid. Naively encoding shared-scale Coulomb–Sturmians in L^2 inflates the block-encoding 1-norm (the non-orthogonal overlap must be Löwdin-inverted, and $S^{-1/2}$ is dense); Avery's *isoenergetic* (potential-weighted) reformulation removes the metric *for atoms* entirely—a metric-free standard eigenproblem whose eigenvalues are the energies [33]. Its 1-norm grows more slowly than the matrix dimension over every basis the paper can compute ($\|M\|_1 \sim K^{0.82}$ fitted on $K = 74$–164, helium), and *increasingly* so: the local slope falls monotonically to 0.766 by $K = 514$, so no window fit is stable and the object is closer to $K^p \log K$ than to a power law. The nuclear diagonal $T^0 = Z \sum_\nu R_\nu$ is the slowest-growing block and the only one derived rather than measured — $Z\sqrt{2K}\ln(K/2)$, asymptotic exponent 1/2 — while neither grouping of the pure-number block T' is superlinear ($K^{0.88}$ full, $K^{0.94}$ off-diagonal). *The substantive finding is that the advantage belongs to the basis-growth rule, not to the construction*: it holds at fixed $l_{\max}$ and weakens under full-shell growth, where the pure-number block reaches $K^{1.07}$ while the total exponent rises only to 0.911 — still sublinear. The cheap rule carries an accuracy floor of 6.4 mHa. *The mechanism of that floor is now pinned, and it is neither angular truncation nor the span*: the metric-free form exists only when the basis scale is locked to the eigenvalue, $\lambda = p_\kappa = \sqrt{-2E}$ — an algebraic identity, since $E = -\lambda^2/2$ is exactly what cancels the L^2 metric — and that lock is not the variational optimum. Solving the *identical* span variationally with the scale freed reaches 1.28 mHa at $K = 130$ against 7.46 mHa locked, and 0.15 mHa of the independently known s-limit at $K = 136$; the price is the whole encoding advantage, $\|\cdot\|_1$ from $K^{0.72}$ to $K^{2.75}$ on the s-only comparison ladder ($K = 21$–136, not the headline window). Metric-free posing and accuracy floor are one fact rather than two, both being the scale lock; the sublinearity is not part of that identity, since it belongs to the basis-growth rule. The floor is moreover a *ground-state* pathology, and Avery's own mechanism explains why: a Goscinskian $1s^2$ configuration pins both electrons to one exponent, where an excited configuration gets two free from $n_a \neq n_b$. Since $\|M\|_1$ does not depend on which root is extracted, every ^1S state costs the same to encode — and at $K = 452$ the ground state sits 4.28× above chemical accuracy while $2\,^1S$ sits 1.08×, the posing cost falling by 4.3×, 3.0× and 2.6× across the first three rungs. (Superseded 2026-09-07: the earlier $K^{0.84}$, "drifts upward", T^0-at-$K^{0.70}$ and T'-superlinear-at-$K^{1.05}$ [retracted 2026-09-08: p60-tprime-superlinear] readings were measured on a truncated radial domain whose error grew with the fit variable.) That split is what decides the molecular case. For molecules the Shibuya–Wulfman metric returns, and the paper measures rather than assumes its cost: the conditioning grows polynomially, with a derived band-limited law, and of its three levers (the gerade sector, large separation, a metric confined to one electron) the sharpest—the gerade sector's basis-independent conditioning—is an *equivalent-center* property, and a symmetry-unique heavy atom reinstates the *growth*, confining that lever to homonuclear-diatomic-like systems. The N-electron 1-norm question is answered in the negative: the atomic sublinearity is a property of the isoenergetic secular matrix's single-center diagonal, not of the basis, and it does not survive two centers.

For the guardrail above this is the operative datum: the method *binds*—H_2^+ at 1.3% s-only underbinding and a minimal-basis H_2 at $E_{\text{tot}} \approx -1.09$ Ha, in a common-scale basis—because the genuine overlap supplies the R-dependence the single-n approximation lacked. What it does not do is recover GeoVac's angular sparsity, which is the trade Paper 58 states as a conservation law.

E. Graph-native FCI for atoms

The atomic FCI paper [34] solves multi-electron atoms directly on the sparse S^3 graph Hamiltonian using Slater–Condon rules [13, 14] with exact R^k radial integrals and

Gaunt coefficients, with Slater determinants indexed as sorted tuples of spin-orbital labels (relational lattice indexing). Helium reaches 0.35% at $n_{\max} = 5$ (grid-based) and 0.19% at $n_{\max} = 7$ with exact rational Slater integrals; lithium 1.07% (at $n_{\max} = 5$) and beryllium 0.90% (at $n_{\max} = 4$, with 487,635 determinants) [34]. The Hamiltonian is sparse (fill fraction 0.68% for He at $n_{\max} = 5$, 0.12% for Li at $n_{\max} = 4$) with a subquadratic nonzero-count growth exponent $\alpha \approx 1.4$–1.5. The paper documents an instructive variational subtlety: a hybrid one-body operator that mixes graph Laplacian off-diagonals (carrying $\kappa = -1/16$) improves He but violates the variational principle for three-electron lithium, so the exact hydrogenic one-body operator is required for monotonic convergence [34]. The dominant remaining error for heteronuclear LiH at $n_{\max} = 3$ is basis truncation, not integral incompleteness—the basis-set superposition error there exceeds the experimental binding energy, so the raw binding number is reported with its counterpoise correction [15] and its honest caveat [34].

F. The graph-concatenation theorem (a second guardrail)

The molecular FCI paper [35] supplies the arc's second proven negative theorem. A topological LCAO architecture that builds a molecular graph by concatenating atom-centered hydrogenic lattices (via Mulliken cross-nuclear attraction, Slater–Condon repulsion, and overlap bridges) is shown to have an intra-atomic graph-Laplacian kinetic energy that is *completely R-independent* by construction: the atomic lattice structure does not change as the nuclei approach. The balanced Hamiltonian therefore produces a monotonically attractive potential-energy surface with no equilibrium geometry—the binding energy decreases smoothly from 0.464 Ha at $R = 1.5$ bohr to 0.042 Ha at $R = 5.0$ bohr, with a virial ratio far from unity [35]. The paper documents a 29-version diagnostic arc that falsifies basis-set superposition error as the geometric driver, rules out orbital-exponent relaxation, and—citing the Sturmian structural theorem of Paper 8—rules out every single-center Sturmian and bond-sphere alternative *of the single-n D-matrix form* as a repair; a single-parameter Fock-weighted kinetic correction partially restores an equilibrium ($R_{\mathrm{eq}} \approx 3.8$ bohr) but cannot fit R_{eq} and D_e simultaneously [22, 35].

Together the two negative theorems—Sturmian structural (shared $SO(4)$ Wigner-D congruence $\Rightarrow$ R-independent eigenvalues, Paper 8) and graph concatenation (R-independent kinetic energy $\Rightarrow$ no equilibrium, the molecular FCI paper)—are guardrails: they explain *why* the natural-geometry hierarchy is necessary. A molecule must be discretized in coordinates where the separation introduces R-dependence (prolate spheroidal, molecule-frame hyperspherical, or composed), because the two obvious shortcuts (one shared-exponent single-center basis in the single-n D-matrix approximation; one concatenated graph) cannot bind in the forms analysed. The shared-exponent leg is the scoped one: the genuine cross-n basis escapes it algebraically (Paper 8 `cor:binding`; Paper 60), at the cost of the qubit-encoding sparsity that motivated the restriction.

VII. THE ALGEBRAIC-FIRST THREAD

A single discipline runs through every level of the hierarchy: replace continuous integration with algebraic evaluation wherever the structure allows, and name what remains. This is the chemistry-arc instance of the framework's prime directive (the discrete angular/selection-rule structure is exact at all levels and must never be approximated; the radial/parametric content is level-dependent and a legitimate target for numerical improvement). Table II catalogues the state of the algebraic registry across the arc.

The pattern is consistent with the foundations-arc taxonomy of Paper 18 [28]. The angular sector is intrinsic (rational, π-free). The diatomic radial seed $e^a E_1(a)$ and the Slater-integral seeds are composition-tier transcendentals tied to a specific integral. The electron–electron cusp is the embedding-tier exchange constant. And the items still in the numerical column—the Z_{eff} screening, the adiabatic curves, the hyperradial coupling—are precisely the places where an algebraic structure has not yet been found, which the project records as an ongoing goal rather than a deficiency: the Neumann expansion (Paper 12) and the split-region Legendre termination (Paper 15) are existence proofs that quadrature walls in this arc *can* dissolve once the right structure is identified [24, 26].

VIII. POSITIONING: A RESEARCH INSTRUMENT

The classical-solver investigation summarized in this synthesis is complete: the natural-geometry hierarchy has been built level by level, its algebraic content catalogued, and its accuracy ceilings characterized. The framework's value proposition is best stated against those ceilings honestly.

A. What the framework offers

The construction is *zero-parameter*: every result in Table I takes as input only nuclear charges, geometry, and a basis resolution—no fitted potentials, no empirical screening constants, no experimental molecular data (the PK pseudopotential of Paper 17 is derived entirely from the atomic core wavefunction) [23, 27]. Angular-momentum selection rules are baked into the basis, so the matrix elements that vanish by symmetry are never

TABLE II. The algebraic-first registry across the chemistry arc. *Algebraic* = closed-form or exact-recurrence from quantum-number labels with no quadrature; *algebraic (one seed)* = exact up to a single tagged transcendental; *numerical* = currently requires quadrature or point-by-point diagonalization, named as a present (not fundamental) feature.

Level / object	Status	Mechanism (paper)
Angular coupling (all levels)	algebraic	Gaunt / Wigner $3j$; selection rules [12, 25, 26]
H_2^+ radial, σ ($m=0$)	algebraic	three-term Laguerre recurrence [23]
H_2^+ radial, π/δ ($m \neq 0$)	algebraic (one seed)	assoc. Laguerre + seed $e^a E_1(a)$ [23]
Prolate spheroidal V_{ee}	algebraic	Neumann expansion, recurrence moments [24]
He hyperradial S, K matrices	algebraic	three-term Laguerre recurrence [25]
Nuclear coupling (Level 4)	algebraic	split-region Legendre, $3j$-triangle termination [26]
Cross-center V_{ne} (balanced)	algebraic	multipole, exact Gaunt termination $L_{\max} = 2l_{\max}$ [29]
Slater F^k integrals (graph-native)	algebraic	exact rational S^3 node overlap [25, 34]
Z_{eff} screening	numerical	Gauss–Legendre quadrature [26, 27]
Adiabatic potential $U(R)$, $\mu(R)$	numerical	point-by-point diagonalization [25]
Hyperradial coupling (Levels 3–4)	numerical	FD / spectral radial solve [25, 26]

computed, and the surviving ones are exact algebraic quantities. The Hamiltonians are structurally sparse, with sub-quadratic nonzero growth [34]. These structural properties—not interpretive claims about the nature of quantum mechanics—are the framework's actual advantages.

B. Where the headline value lives

The arc's strongest computational payoff is not the classical ground-state energies but the qubit Hamiltonians they imply. The angular sparsity that this synthesis documents at the integral level is the structural reason the composed-geometry qubit encodings achieve favorable Pauli-term scaling for variational quantum simulation; that result, its scaling exponents, and its molecular benchmarks belong to the quantum-computing arc (Group 4) and are referenced here, not re-derived. Paper 58 [30] sharpens the scope of that payoff: the l-selection component of the sparsity is an atomic-sector property, exact for the composed encoding precisely because it evaluates per center, so the favorable scaling characterizes small systems evaluated in the atomic sector rather than genuine multi-center integration. This synthesis establishes the chemistry-side foundation of that payoff: the natural-geometry hierarchy, the algebraic-first integral structure, and the two negative theorems that justify the composed encoding.

C. The benchmarking discipline

Where the framework is compared to other methods, the comparison is to the strongest available baseline, and unfavorable comparisons are stated plainly. The composed potential-energy surfaces carry 5–19.4% bond-length error, well above production quantum-chemistry accuracy; the honest reading is that the framework is a computationally principled *alternative* offering sparsity, scaling, and structural insight, not a replacement for correlated electronic-structure methods. The papers report the proven negatives—the Sturmian and concatenation theorems, the PK ceiling, the W1e wall, the failed PK alternatives and cusp treatments—as a first-class part of the corpus, because the characterization of what natural geometry *cannot* do is as load-bearing as what it can [22, 27, 35].

IX. OPEN QUESTIONS AND OUTLOOK

Three structural questions remain open at the close of the classical-solver arc.

a. The composed accuracy ceiling (W1e). The composed architecture saturates near 5–6% bond-length error at its $l_{\max} = 2$ operating point. Whether a composition rule could close W1e without sacrificing the 144× angular compression—i.e. without reverting to the exact-but-expensive Level-4N encoding—was the central open chemistry question of the classical-solver arc; Paper 58 answers it in the negative and gives the reason [30]. The l-selection sparsity is exact only in the atomic sector—it is the abelian residue that survives genuine two-center evaluation—so closing the bond requires forming the two-center orbital products that destroy it. W1e is thereby relocated from the correlation treatment (a DMRG cross-check reproduces FCI bit-identically without binding) to the Hamiltonian specification itself, and the sparsity-versus-binding trade is a conservation law rather than a missing integral. What remains open is quantitative: how much of the residual a genuine-integral construction recovers across a range of systems, of which NaH (+4.8%) is the first data point.

b. Algebraic replacement of the remaining quadrature. The numerical column of Table II—Z_{eff} screening, the adiabatic potential curves, the hyperradial coupling—is the standing target of the algebraic-first program. The existence proofs (Neumann V_{ee}, split-region Legendre termination) show such replacements are achievable in principle; whether the adiabatic curves

$\mu(R)$, which Paper 13 records as genuinely transcendental in R, admit an algebraic-implicit form (defined by a polynomial equation over a known coefficient ring) is open.

c. Second-row and polyatomic extension. The balanced coupled Hamiltonian hit an orthogonalization wall for second-row systems such as NaH [29]; the genuine two-center construction of Paper 58 crosses it (NaH binds at +4.8%), at the cost of the sparsity [30]. The polyatomic H_2O construction validates only bond–bond coupling, with bond–lone and lone–lone couplings unphysical at the composed level [27]; extending the genuine-integral route past two centers is the open polyatomic question—now sharpened by Paper 59 from an algebra question to a transcendence one: the third center raises the genus from zero to one, the resulting connection is irreducible, and the finite closed form is open [31]. Both remain named boundaries of the current encoding rather than refutations of the hierarchy.

None of these open questions destabilizes the structural results of the arc. The natural-geometry hierarchy is established level by level; the algebraic-first content is catalogued; the two negative theorems are proven; and the accuracy ceilings are characterized. The open questions are frontier questions about how far the framework's sparsity can be pushed before its accuracy ceiling binds, not foundational concerns.

X. CONCLUSION

The Group 2 quantum-chemistry arc of GeoVac is the systematic construction of the natural-geometry hierarchy for electronic structure. Each level is the coordinate system in which a given system separates: S^3 for atoms [21], the prolate spheroid for H_2^+ (0.0002%) [23], hyperspherical coordinates for He (0.022% variational raw / 0.004% non-variational cusp, 0.19% graph-native) [25], molecule-frame hyperspherical coordinates for H_2 (96.0% of D_e) [26], and composed fiber bundles for core–valence and polyatomic systems (5.3%, 11.7%, 19.4% bond-length error for LiH, BeH_2, H_2O) [27]. The arc is held together by an algebraic-first discipline—exact angular coupling, algebraic Neumann V_{ee} [24], Laguerre recurrences, and $3j$-terminating Legendre expansions [26]—and is bounded by two structural negative theorems: a single-center shared-p_0 Sturmian basis cannot bind a molecule *in the single-n D-matrix approximation* [22] (the genuine cross-n basis does bind, algebraically, at the cost of the sparsity—[33]), and a concatenated atom-centered graph has no equilibrium [35]. Those theorems are why the hierarchy is the route this arc took; the first is a scoped guardrail rather than an unconditional no-go.

The framework is a zero-parameter, structurally sparse research instrument whose accuracy ceilings are now characterized and whose strongest downstream payoff—qubit-Hamiltonian sparsity for quantum simulation—lives in the quantum-computing arc. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps; the framework's concrete advantages do not depend on which description one regards as fundamental, and that interpretive question is left open.

ACKNOWLEDGMENTS

This synthesis paper consolidates work across the Group 2 quantum-chemistry arc of the GeoVac project. The author thanks the broader open-source computational quantum-chemistry community for the libraries (NumPy, SciPy, sympy, mpmath, PySCF) that make this work possible. The framework's papers and code are publicly archived on Zenodo with permanent DOIs; this synthesis is a project-archive document and is not formatted for journal submission.

[1] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).
[2] D. R. Bates, K. Ledsham, and A. L. Stewart, "Wave functions of the hydrogen molecular ion," Philos. Trans. R. Soc. Lond. A **246**, 215 (1953).
[3] C. Flammer, *Spheroidal Wave Functions* (Stanford University Press, Stanford, 1957).
[4] C. Neumann, *Theorie der Elektricitäts- und Wärme-Vertheilung in einem Ringe* (Halle, 1864).
[5] H. M. James and A. S. Coolidge, "The ground state of the hydrogen molecule," J. Chem. Phys. **1**, 825 (1933).
[6] W. Kołos and L. Wolniewicz, "Improved theoretical ground-state energy of the hydrogen molecule," J. Chem. Phys. **49**, 404 (1968).
[7] T. Kato, "On the eigenfunctions of many-particle systems in quantum mechanics," Commun. Pure Appl. Math. **10**, 151 (1957).
[8] V. Aquilanti, S. Cavalli, and G. Grossi, "Hyperspherical coordinates for molecular dynamics by the method of trees," J. Chem. Phys. **85**, 1362 (1986).
[9] J. Avery, *Hyperspherical Harmonics: Applications in Quantum Theory* (Kluwer Academic, Dordrecht, 1989).
[10] J. E. Avery and J. S. Avery, *Generalized Sturmians and Atomic Spectra* (World Scientific, Singapore, 2006).
[11] T. Shibuya and C. E. Wulfman, "Molecular orbitals in momentum space," Proc. R. Soc. Lond. A **286**, 376 (1965).
[12] L. C. Biedenharn and J. D. Louck, *Angular Momentum in Quantum Physics* (Addison-Wesley, Reading, MA, 1981).
[13] E. U. Condon and G. H. Shortley, *The Theory of Atomic Spectra* (Cambridge University Press, Cambridge, 1935).
[14] J. C. Slater, *Quantum Theory of Atomic Structure*, Vols. I and II (McGraw-Hill, New York, 1960).

[15] S. F. Boys and F. Bernardi, "The calculation of small molecular interactions by the differences of separate total energies," Mol. Phys. **19**, 553 (1970).

[16] K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure IV: Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979).

[17] J. C. Phillips and L. Kleinman, "New method for calculating wave functions in crystals and molecules," Phys. Rev. **116**, 287 (1959).

[18] L. N. Trefethen, *Spectral Methods in MATLAB* (SIAM, Philadelphia, 2000).

[19] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[20] J. Loutey, "The Geometric Atom: Atomic Structure as a Packing Problem," GeoVac Paper 1 (2026).

[21] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[22] J. Loutey, "Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis," GeoVac Papers 8–9 (2026).

[23] J. Loutey, "The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices," GeoVac Paper 11 (2026).

[24] J. Loutey, "Algebraic Two-Electron Integrals on the Prolate Spheroidal Lattice," GeoVac Paper 12 (2026).

[25] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," GeoVac Paper 13 (2026).

[26] J. Loutey, "The Level 4 Natural Geometry: Two-Center Two-Electron Molecules in Molecule-Frame Hyperspherical Coordinates," GeoVac Paper 15 (2026).

[27] J. Loutey, "Composed Natural Geometries for Core–Valence Diatomics: LiH and BeH^+ from Discrete Graph Topology," GeoVac Paper 17 (2026).

[28] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[29] J. Loutey, "Toward PK-Free Molecular Hamiltonians: Two-Center Sturmian Integrals and Coupled Composition," GeoVac Paper 19 (2026).

[30] J. Loutey, "Angular Sparsity Is an Atomic-Sector Property: The Abelian Residue of Multi-Center Bonding," GeoVac Paper 58 (2026).

[31] J. Loutey, "The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory," GeoVac Paper 59 (2026).

[32] J. Loutey, "The Bessel-Moment Period Algebra of the Three-Center Coulomb Integral: Modular Structure and Cosmic-Galois Placement," GeoVac Paper 61 (2026).

[33] J. Loutey, "The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric: A Quantum-Algorithm Analysis, Metric-Free for Atoms and a Conditioning Frontier for Molecules," GeoVac Paper 60 (2026).

[34] J. Loutey, "Full Configuration Interaction on a Sparse Graph Hamiltonian: Multi-electron Atoms via Relational Lattice Indexing," GeoVac FCI (atoms) Paper (2026).

[35] J. Loutey, "Topological Full Configuration Interaction for Heteronuclear Diatomics: LiH as a Benchmark," GeoVac FCI (molecules) Paper (2026).